

# Network Forensic Report

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## PCAP Network Packet Capture Analysis

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**On the Date 2/18/2025**

This report provides the details of the forensic analysis performed on the network capture file "XYZ.pcap."

## Table of Contents

|                                              |    |
|----------------------------------------------|----|
| 1. Executive Summary .....                   | 3  |
| 2. Introduction .....                        | 4  |
| 2.1 Network Capture File details .....       | 4  |
| 2.2 Network Components Identified.....       | 7  |
| 3. Methodology .....                         | 9  |
| 3.1 Tools Used .....                         | 9  |
| 3.2 Steps Involved.....                      | 10 |
| 3.3 Handling Data .....                      | 13 |
| 4. Detailed Findings .....                   | 15 |
| 4.1 Important Network Players .....          | 15 |
| 4.2 Network Structure .....                  | 16 |
| 4.3 Activity Timeline for 192.168.15.4 ..... | 17 |
| 4.4 Background Evidence .....                | 18 |
| Supporting Evidence Presented .....          | 20 |
| 5. Conclusions .....                         | 23 |
| 6. Appendix A – List of figures .....        | 24 |
| 7. Self-Review.....                          | 26 |

## 1. Executive Summary

This report contains the forensic analysis conducted for the sole purpose of determining whether a student from Chemistry 109 in the XYZ school was responsible for sending harassing emails one of which was sent with a web-based service called willselfdestruct.com and the other the primary threat focused on to Lily Tuckridge, the teacher of said class and department. The investigation primarily focuses on the network traffic captured using a network sniffer placed on the ethernet port at the school, from the IP address of the dormitory 140.247.62.34, which was linked to the harassing emails. For detailed review and investigation, the following pcap file called XYZ.pcap was reviewed using software such as Wireshark and Network Miner. The students of Chemistry 109 class are, Amy Smith, Burt Greedom, Tuck Gorge, Ava Book, Johnny Coach, Jeremy Ledvkin, Nancy Colburne, Tamara Perkins, Esther Pringle, Asar Misrad, Jenny Kant, and in the dorm there's Alice, Barbara, Candice, and lastly Barbara's boyfriend who installed the router and that concludes all the main suspects.

1. The two routers (00:1d:d9:2e:4f:60, 00:1d:d9:2e:4f:61) were monitored directly by the network sniffer, and all information and packets that were monitored were with the help of that in order to find the harassment activities.
2. DNS response using frame contains "willselfdestruct" && dns gives us conclusive information about all the searches that were made to that website, within that time.
3. The suspects device runs on a MacOS with (User-Agent: Mozilla/5.0 (Macintosh; PPC Mac OS X)), indicating the use of an apple device for these activities.
4. The email ID jcoachj@gmail.com and MAC address 00:17:f2:e2:c0:ce strongly implicate "jcoachj" as the primary suspect. This was done by filtering based on the public ip to find connection to the private ip, then coming across the MAC address they provide which was the same throughout the packets ensuring the connection to the private IP, and with the TCP stream confirming the device used.
5. We made use of NetworkMiner and Wireshark to analyse the network traffic, to get the information sufficient to validate our findings.
6. Traffic to sendanonymousemail.net Frame number 80614 and WillSelfDestruct.com Frame number 83601 originated from 192.168.15.4, confirming jcoachj's device as the source.
7. Filtering on keywords such as teacher and gmail revealed Frame number 74920 a google query containing "can I go to jail for harassing my teacher", and Frame number 79715 a cleartext cookie containing the cookie pair of jcoachj.
8. The DNS requests for harassment domains was resolved to the XYZ campus network, which means any alibi of it being off-campus activity is completely improbable.
9. Another ID was found connecting to the device jcoachj was using, under the alias amy789smith indicating device sharing or an attempt to get the device from the original owner, both this individual and later mentioned elishevet have shown no involvement in this harassment attempt, but certain aspects have been monitored with the idea of further proofing the involvement of jcoachj.
10. Consider that 73000~ packets out of the 94000~ were strictly communication between MacOS device used by jcoachj and the dorm router.
11. Evidently both the device and email match a student in the Chemistry 109 class, Johnny Coach, all evidence that is pointing to him has been recorded in the supporting evidence.

12. All evidence provided complies with forensics standards for chain of custody.

## 2. Introduction

### 2.1 Network Capture File details

The extracted PCAP network capture file XYZ.pcap has the forensic parameters as given below. The evidence for these details is provided in Figure 1 extracted from Wireshark version 4.2.5. (Git v4.2.5 packaged as 4.2.5-1)

|                            |                                                             |
|----------------------------|-------------------------------------------------------------|
| Capture Length             | 54670237 bytes (Approximately 56 MB)                        |
| Format                     | Wireshark/tcpdump/... - pcap                                |
| Packet Size Limit          | 4096 bytes                                                  |
| First Packet               | Date - 2008-07-21 Time - 18:51:07                           |
| Last Packet                | Date - 2008-07-21 Time - 23:13:47                           |
| Elapsed Time               | 04:22:39 -<br>4 Hours 22 Minutes 39 Seconds (15759 Seconds) |
| Total Packets              | 94, 410 Packets                                             |
| Average Packets Per Second | 6.0 Packets Per Second                                      |
| Average Packet Size        | 579 bytes                                                   |
| Average bytes Per Second   | 3468 bytes Per Second                                       |

Table 1 – Forensic Parameters of the PCAP File

#### Computed Hashes – XYZ.pcap.tar.gz

| Hash Type | Hash Value                                                       | Verified By |
|-----------|------------------------------------------------------------------|-------------|
| MD5       | b778b653060ef14e905c2d1c185dc1a6                                 | Hash Rat    |
| SHA1      | fbaca21fdflcaea6b259b66bec0aff4d5d2f910f                         | Hash Rat    |
| SHA256    | 8089c69922dda4f507de01599e376017f6a6b0c6506c2968f88012670ff27e57 | Hash Rat    |

Table 2. Hashes of the Archive Version of PCAP File (Verified)

#### Computed Hashes – XYZ.pcap

| Hash Type | Hash Value                                                       | Verified By              |
|-----------|------------------------------------------------------------------|--------------------------|
| MD5       | 9981827f11968773ff815e39f5458ec8                                 | Network Miner & Hash Rat |
| SHA1      | 65656392412add15f93f8585197a8998aaeb50a1                         | Wireshark & Hash Rat     |
| SHA256    | 2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb | Wireshark & Hash Rat     |

Table 3. Hashes of the PCAP File (Verified)

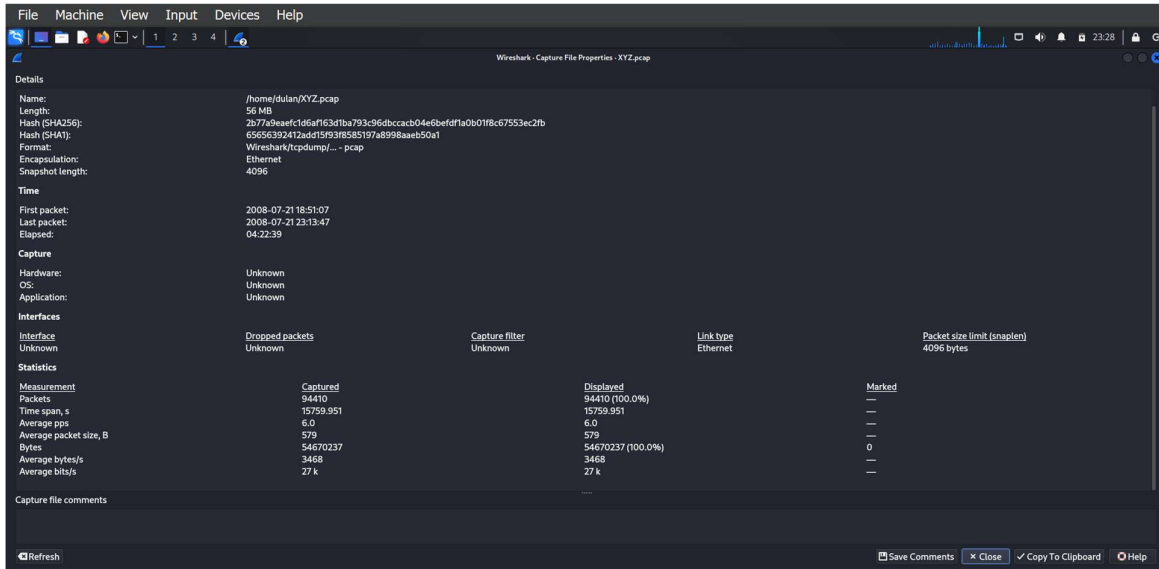


Figure 1. Packet Capture Summary from Wireshark Version 4.2.5 Including Sha1 and Sha256 Hashes

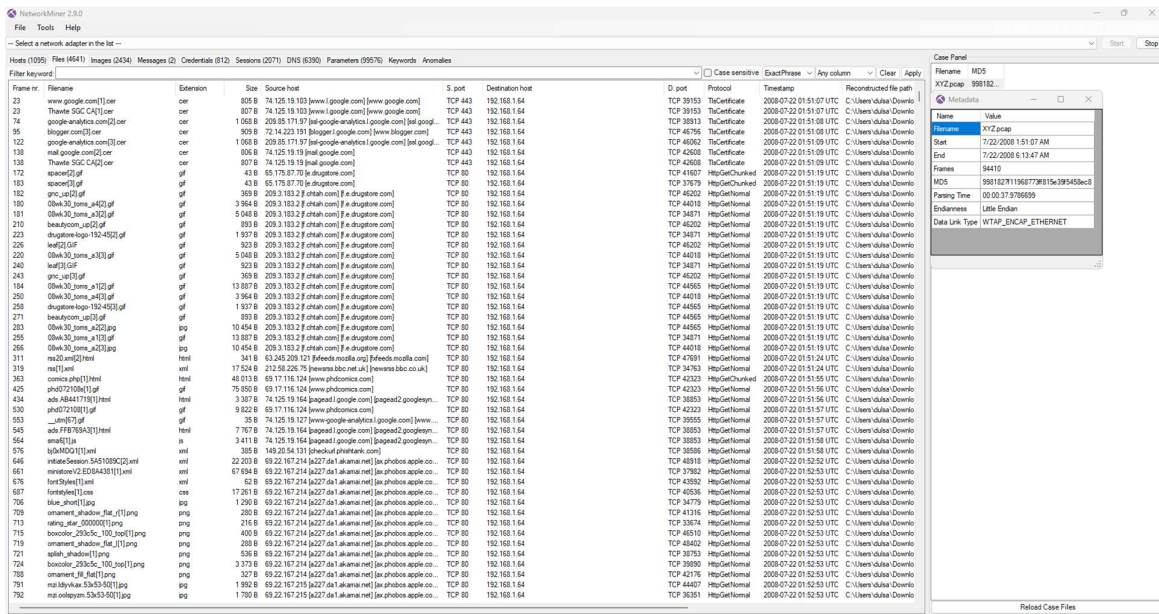


Figure 2. Metadata of the PCAP File from Network Miner Version 2.9.0, Including MD5 Hash Value.

```
Dulan Nithila [yanaradchi's Kali Linux (CSE Lab 8 & DFAEH Practical Lab) [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
dulan@kali -
File Actions Edit View Help
[dulan@kali:~]$ hashrat -version
version: 1.28
[dulan@kali:~]$
[dulan@kali:~]$ hashrat -md5 XYZ.pcap
hash= md5:9051827f11968773f815d39f5a58ec8 type='file' mode='100777' uid='1000' gid='1000' size='56180821' mtime='1738558184' inode='1434756' path='XYZ.pcap'
[dulan@kali:~]$
[dulan@kali:~]$ hashrat -md5 XYZ.pcap.tar.gz
hash= md5:57780693806ef1ee00c201c183dca8 type='file' mode='100664' uid='1000' gid='1000' size='39445918' mtime='1739374849' inode='43907' path='XYZ.pcap.tar.gz'
[dulan@kali:~]$
[dulan@kali:~]$ hashrat -sha1 XYZ.pcap
hash= sha1:45836392412d0d15f93f8305197d998baa658a1 type='file' mode='100777' uid='1000' gid='1000' size='56180821' mtime='1738558184' inode='1434756' path='XYZ.pcap'
[dulan@kali:~]$
[dulan@kali:~]$ hashrat -sha1 XYZ.pcap.tar.gz
hash= sha1:f0aca21fd0f1caeb0259606bcb8ffdc5d7910f type='file' mode='100664' uid='1000' gid='1000' size='39445918' mtime='1739374849' inode='43907' path='XYZ.pcap.tar.gz'
[dulan@kali:~]$
[dulan@kali:~]$ hashrat -sha256 XYZ.pcap
hash= sha256:2077a9aeef1d6af163d1ba791c96dbccacbb4e5b0efdf1e0b1f8c67551ec2fb type='file' mode='100777' uid='1000' gid='1000' size='56180821' mtime='1738558184' inode='1434756' path='XYZ.pcap'
[dulan@kali:~]$
[dulan@kali:~]$ hashrat -sha256 XYZ.pcap.tar.gz
hash= sha256:1889c69922d6aef07de01599e776817faa0bc506c2968f8012670ff27e57 type='file' mode='100664' uid='1000' gid='1000' size='39445918' mtime='1739374849' inode='43907' path='XYZ.pcap.tar.gz'
[dulan@kali:~]$
```

Figure 3. Verifying The Hashes of The Pcap File and The Archive Version Using Hash Rat

## 2.2 Network Components Identified

According to the capture summary as provided by Wireshark version 4.2.5, there are 17 distinct Ethernet components. They were determined using the Ethernet Endpoints listed under Statistics as below.

| No | MAC Address       | MAC Address with Name Resolution         | IP Address     | Vendor                          | Device    | OS                    |
|----|-------------------|------------------------------------------|----------------|---------------------------------|-----------|-----------------------|
| 1  | 00:0a:95:69:38:cc | Apple_69:38:cc                           | 192.168.1.5    | Apple, Inc.                     | -         | -                     |
| 2  | 00:14:d1:44:a0:f1 | TRENDnet_44:a0:f1                        | 192.168.15.5   | TRENDnet, Inc                   | -         | -                     |
| 3  | 00:16:cb:b4:a3:f8 | Apple_b4:a3:f8                           | 192.168.15.8   | Apple, Inc.                     | -         | -                     |
| 4  | 00:17:f2:e2:c0:ce | Apple_e2:c0:ce                           | 192.168.15.4   | Apple, Inc.                     | Macintosh | Intel Mac OS X 10.5.4 |
| 5  | 00:17:f2:e2:c0:cf | Apple_e2:c0:cf                           | -              | Apple, Inc.                     | -         | -                     |
| 6  | 00:1b:63:f1:8a:6e | Apple_f1:8a:6e                           | 192.168.15.2   | Apple, Inc.                     | -         | -                     |
| 7  | 00:1c:b3:79:00:31 | Apple_79:00:31                           | 10.0.1.5       | Apple, Inc.                     | -         | -                     |
| 8  | 00:1d:6b:99:98:68 | ARRISGroup_99:98:68 (Commscope_99:98:68) | 192.168.1.254  | ARRIS Group, Inc.               | Router    | -                     |
| 9  | 00:1d:d9:2e:4f:60 | HonHaiPrecis_2e:4f:60                    | 192.168.15.1   | Hon Hai Precision Ind. Co.,Ltd. | Router    | -                     |
| 10 | 00:1d:d9:2e:4f:61 | HonHaiPrecis_2e:4f:61                    | 192.168.1.64   | Hon Hai Precision Ind. Co.,Ltd. | Router    | -                     |
| 11 | 00:1f:f3:5a:77:9b | Apple_5a:77:9b                           | 169.254.90.183 | Apple, Inc.                     | -         | -                     |
| 12 | 01:00:5e:00:00:02 | IPv4mcast_02                             | -              | -                               | -         | -                     |
| 13 | 01:00:5e:00:00:09 | IPv4mcast_09                             | -              | -                               | -         | -                     |
| 14 | 01:00:5e:00:00:16 | IPv4mcast_16                             | -              | -                               | -         | -                     |
| 15 | 01:00:5e:00:00:fb | IPv4mcast_fb                             | -              | -                               | -         | -                     |
| 16 | 01:00:5e:7f:ff:fa | IPv4mcast_7f:ff:fa                       | -              | -                               | -         | -                     |
| 17 | ff:ff:ff:ff:ff:ff | Broadcast                                | -              | -                               | -         | -                     |

Table 4. Ethernet Component List in The PCAP File

Based on the details of the statistics report generated by Wireshark , it appears that the devices with MAC Addresses Apple\_e2:c0:ce, IPv4mcast\_02, IPv4mcast\_09, IPv4mcast\_16, IPv4mcast\_7f:ff:fa, IPv4mcast\_fb, TRENDnet\_44:a0:f1 were not significantly involved in the conversations tracked on the network capture file and hence can be regarded irrelevant for the

remaining part of the investigation. The device with a name resolution of Apple\_b4:a3:f8 and with the MAC address 00:17:f2:e2:c0:ce was found to have the IP address 192.168.15.4 which was our suspicious individual. He accessed 69.80.225.91 and 69.25.94.22 (Destination IP Addresses) from his device through the router with the name resolution as Broadcast with an IP address of 140.247.62.34 and MAC address of 00:1d:d9:2e:4f:60 . This could potentially imply that the device was actively communicating with external servers, indicating possible unauthorized access or data exfiltration. Based on the evidence of connections established between 192.168.15.4 and the destination IP addresses 69.80.225.91 and 69.25.94.22, this report concludes that the device was engaged in outbound network activity, which may suggest suspicious or unauthorized behavior.

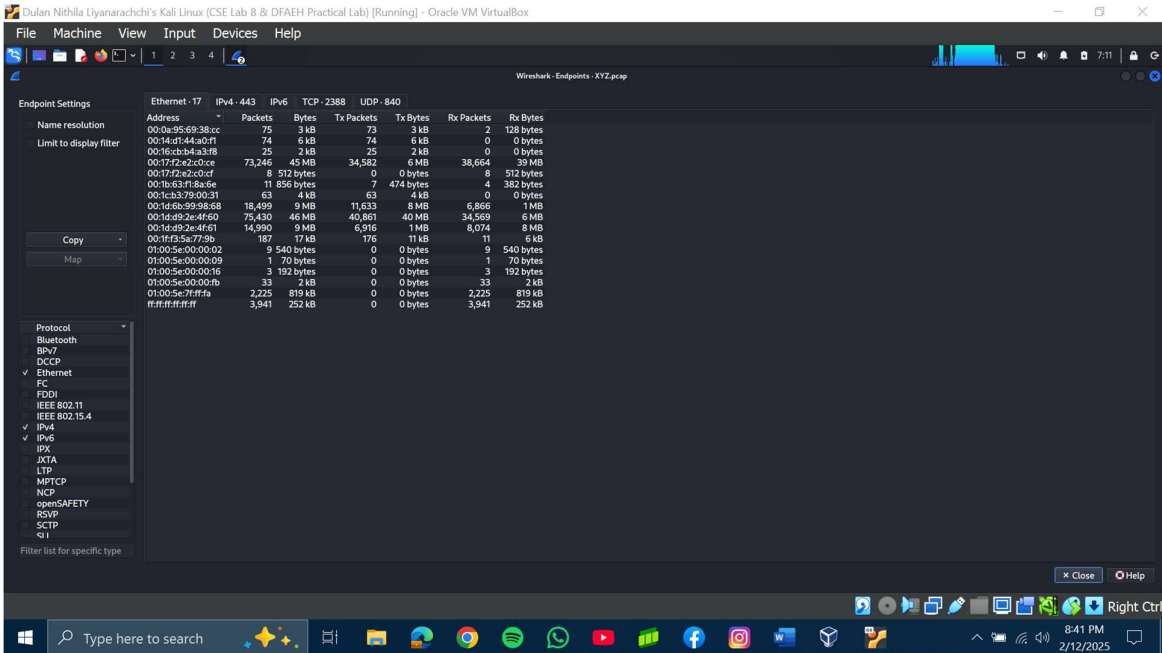


Figure 4. Ethernet Endpoint List under The Statistics Tab in Wireshark

### 3. Methodology

#### 3.1 Tools Used

The analysis analysed the contents using network forensic tools such as Wireshark Version 4.2.5. (Git v4.2.5 packaged as 4.2.5-1) executing under Kali Linux 64 bit on Single CPU with 4 GB RAM in Virtual Box and Wireshark Version.4.4.3 on a separate Windows device running on Windows 11, with 16gb RAM and a 11<sup>th</sup> Gen Intel(R) Core(TM) i7-11370H @ 3.3GHz, with an RTX 3050 Laptop GPU. The findings were later cross verified on another system using Network Miner Version 2.9.0 executing on Windows 11 platform with 16gb RAM and a 11<sup>th</sup> Gen Intel(R) Core(TM) i7-11370H @ 3.3GHz, with an RTX 3050 Laptop GPU. The analysis revealed the internal IP which were used to access to willselfdestruct.com (192.168.15.4) and the external which send the response to the login request(192.168.1.254) through DNS.

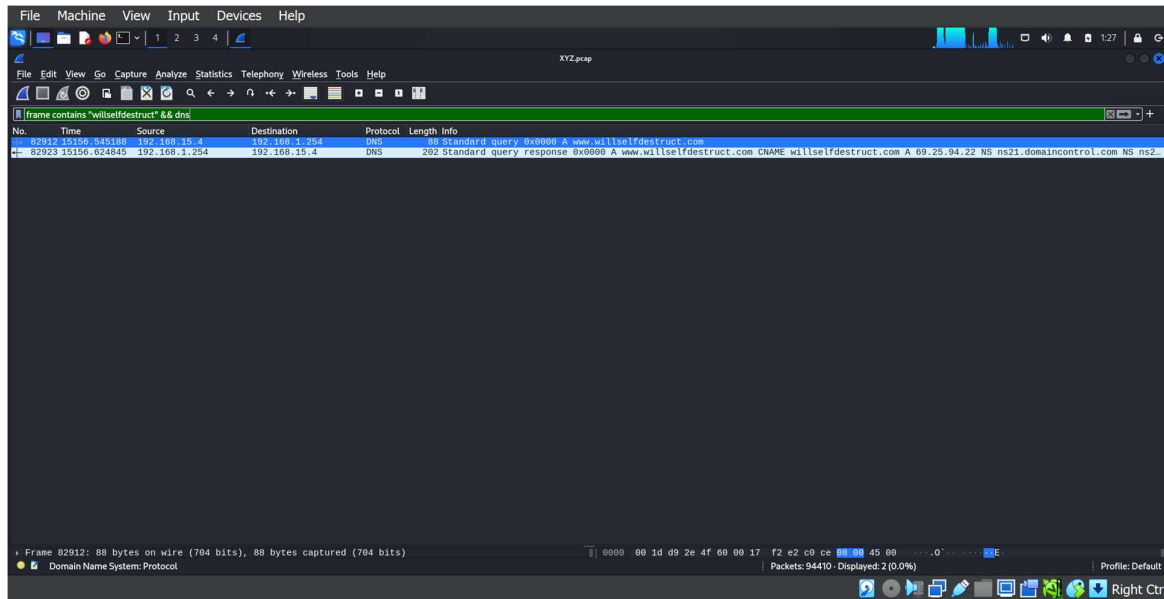


Figure 5. Filtering The Needed Data Using Commands in Wireshark

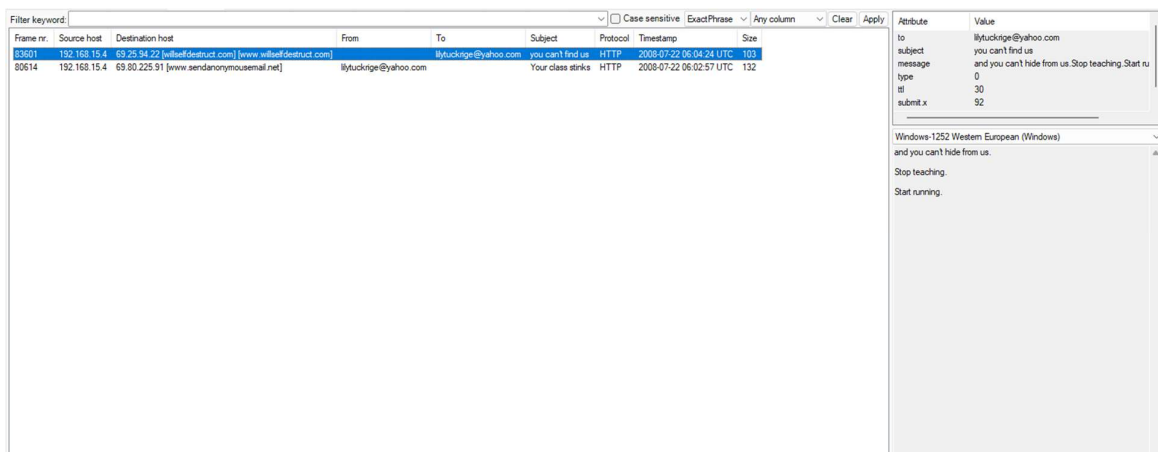


Figure 6. Filtering The Needed Data Using Tools in Network Miner

## 3.2 Steps Involved

The network traffic archive XYZ.pcap was opened in Wireshark v.4.4.3 and Network Miner v2.9.0 for this analysis. We used the following Cryptographic hashes; MD5, SHA1, SHA256 to ensure file integrity prior to giving it an examination. Firstly, by extracting the Private IP with the use of the content mentioned in the background evidence we can filter based on `ip.addr==192.168.15.4` and check the statistics tab provided by Wireshark to get the conversations and endpoints, specifically in endpoints we can see that there's 73,197 or so packets, while the entire pcap file contains around 94,410 total.

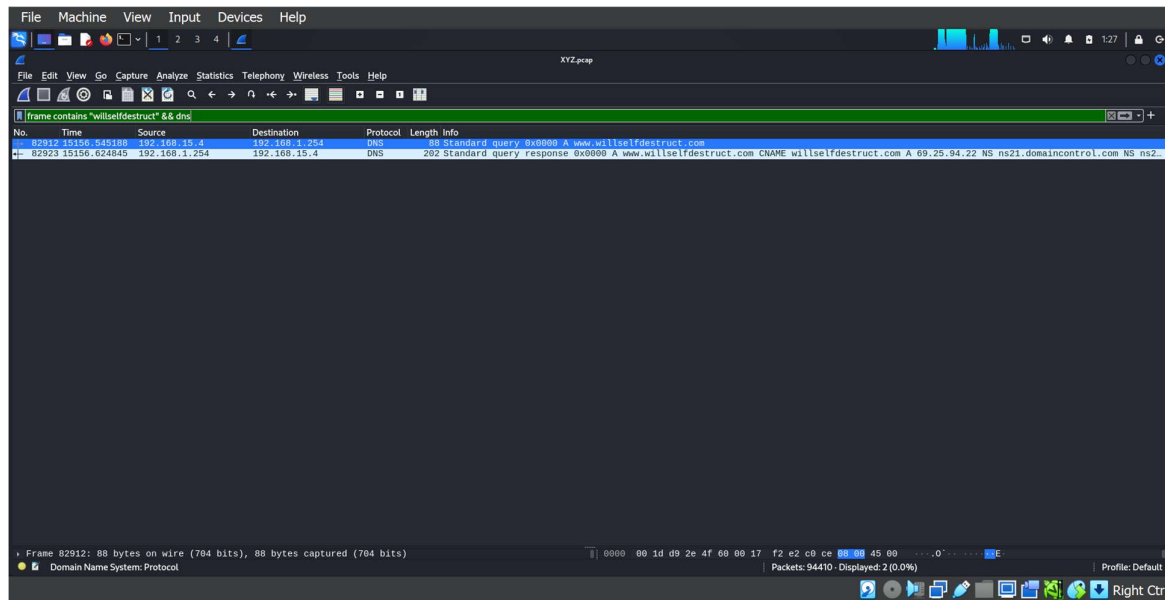


Figure 7. Filtering The Needed Data Using Commands in Wireshark

Filtering steps taken include; `(eth.addr == 00:17:f2:e2:c0:ce)` for activity on MAC Address, `(ip.addr == 192.168.15.4 && http)` activity of the HTTP transactions in correlation to the IP address, keyword searches such as `(frame contains "gmail.com")`, and `http.host contains "willselfdestruct"`, combining filters like `(ip.addr == 192.168.15.4 && eth.addr == 00:17:f2:e2:c0:ce)` then following the TCP stream shown in Figure 9. relates to the specific device used, `(frame contains "lily")` shows those packets set to Lily Tuckridge, `(eth.addr == 00:17:f2:e2:c0:ce && frame contains "search_result")` showing all the searches done by the specific device, and `(eth.addr == 00:17:f2:e2:c0:ce && frame contains "GET /mail")` focusing primarily on the individual and attempting to find their cookie session of their email, focusing entirely on the cookie pairs of the specific MAC address. `(eth.addr == 00:17:f2:e2:c0:ce && http.cookie_pair)`.

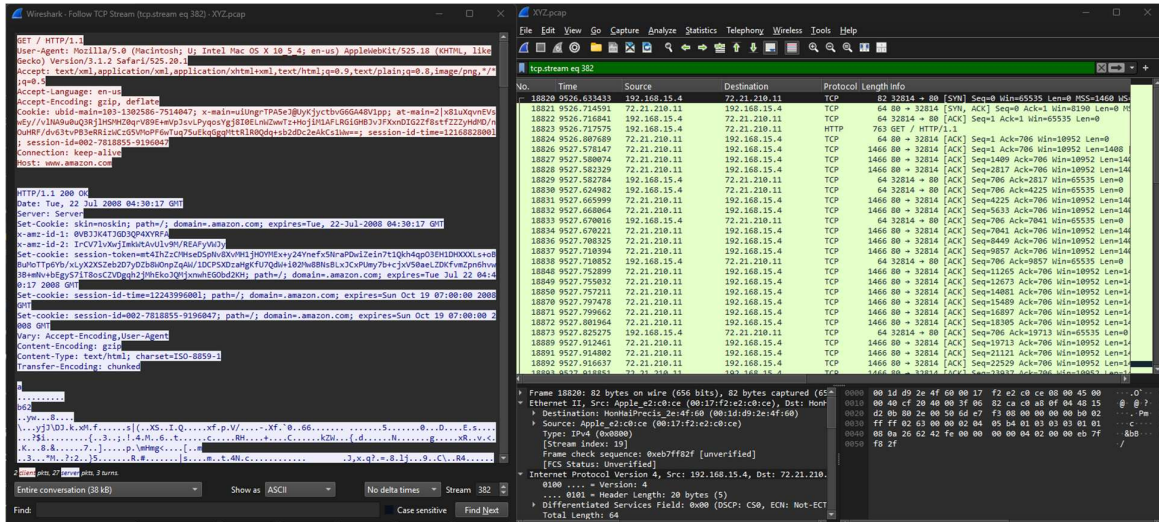


Figure 8. Correlating The MAC Address and IP Address to The Device

Key frame numbers to take note of, 74920 Google search query “can I go to jail for harassing my teacher”, 79715 a clear text gmail cookie of (jcoachj@gmail.com) which was tied to the MAC address of the individual, another one to use is 78967 containing another login session done by jcoachj tied down to the MAC address, 80614, and 83601 respectively containing both threat mails sent to Lily Tuckridge connected to the same IP.

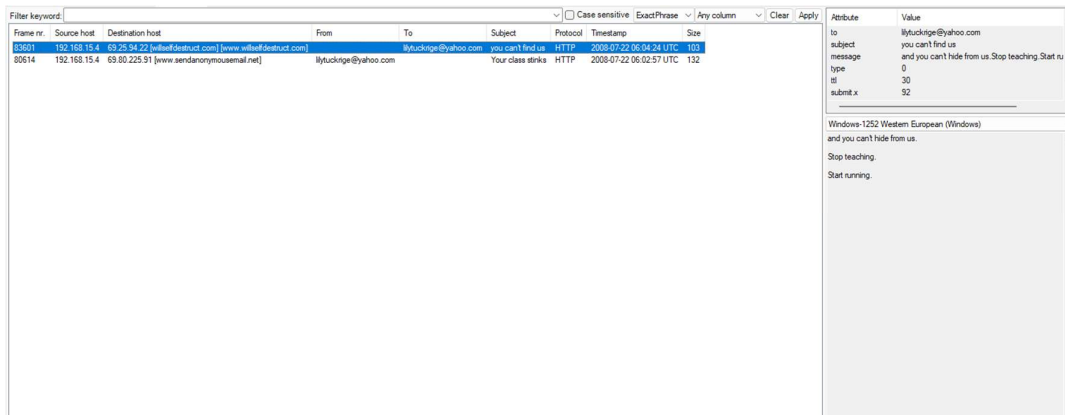


Figure 9. Filtering The Needed Data Using Tools in Network Miner.

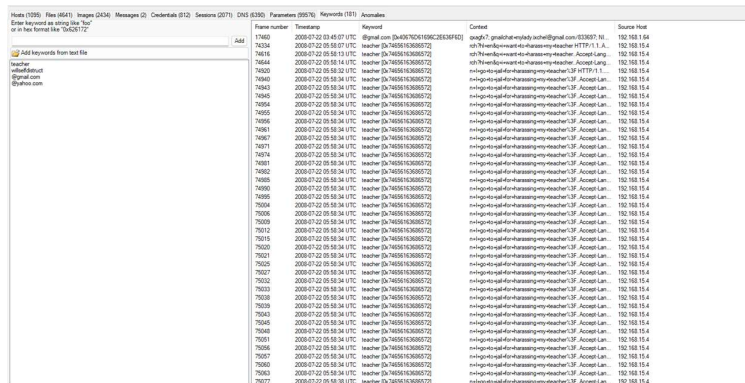


Figure 10. Detailing Packet 74920

| Field                               | Details                                                                                                                   |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Packet Number                       | 74920                                                                                                                     |
| Time                                | 14845.565305                                                                                                              |
| Source                              | 192.168.15.4 (MAC: 00:17:f2:e2:c0:ce, Apple Inc.)                                                                         |
| Destination                         | 209.73.187.220 (MAC: 00:1d:d9:2e:4f:60, Hon Hai Precision Ind. Co.)                                                       |
| Protocol                            | HTTP                                                                                                                      |
| Length                              | 1130 bytes (9040 bits)                                                                                                    |
| Info                                | GET<br>/search/search_result;_ylt=A9FJui4Od4VIL5QANivD7BR.;_ylv=3?p=can+I+go+to+jail+for+harassing+my+teacher%3F HTTP/1.1 |
| Frame Details                       | 1130 bytes on wire (9040 bits), 1130 bytes captured (9040 bits)                                                           |
| Ethernet II                         |                                                                                                                           |
| Destination MAC                     | HonHaiPrecis_2e:4f:60 (00:1d:d9:2e:4f:60)                                                                                 |
| Source MAC                          | Apple_e2:c0:ce (00:17:f2:e2:c0:ce)                                                                                        |
| Type                                | IPv4 (0x0800)                                                                                                             |
| Stream Index                        | 19                                                                                                                        |
| FCS                                 | 0xb04fec5f (Unverified)                                                                                                   |
| Internet Protocol                   |                                                                                                                           |
| Src IP                              | 192.168.15.4                                                                                                              |
| Dst IP                              | 209.73.187.220                                                                                                            |
| Transmission Control Protocol (TCP) |                                                                                                                           |
| Src Port                            | 35650                                                                                                                     |
| Dst Port                            | 80                                                                                                                        |
| Seq                                 | 1                                                                                                                         |
| Ack                                 | 1                                                                                                                         |
| Len                                 | 1060                                                                                                                      |
| Hypertext Transfer Protocol (HTTP)  |                                                                                                                           |
| Request Method                      | GET                                                                                                                       |
| URI                                 | /search/search_result;_ylt=A9FJui4Od4VIL5QANivD7BR.;_ylv=3?p=can+I+go+to+jail+for+harassing+my+teacher%3F                 |
| Version                             | HTTP/1.1                                                                                                                  |

Table 5. Incriminating Evidence Packet 74920

### 3.3 Handling Data

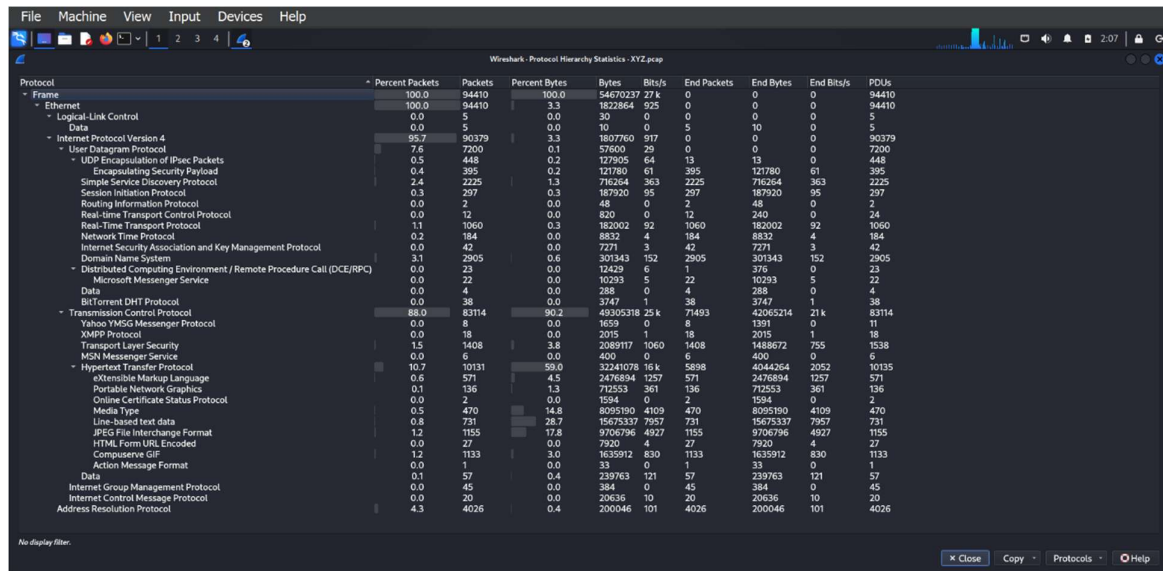


Figure 11. Protocol Hierarchy captured with Wireshark Version 4.2.5.

| Protocol                             | Percent Packets | Packets | Percent Bytes | Bytes    | Bits/s   | End Packets | End Bytes | End Bits/s | PDUs  |
|--------------------------------------|-----------------|---------|---------------|----------|----------|-------------|-----------|------------|-------|
| Frame                                | 100.00          | 94410   | 100.00        | 54670237 | 27751.48 | 0           | 0         | 0.00       | 94410 |
| Ethernet                             | 100.00          | 94410   | 3.33          | 1822854  | 925.31   | 0           | 0         | 0.00       | 94410 |
| Logical-Link Control                 | 0.0053          | 5       | 0.00004       | 20       | 0.01     | 0           | 0         | 0.00       | 5     |
| Data                                 | 0.0053          | 5       | 0.00002       | 10       | 0.0051   | 5           | 10        | 0.0051     | 5     |
| Internet Protocol Version 4          | 95.73           | 90379   | 3.31          | 1807760  | 917.65   | 0           | 0         | 0.00       | 90379 |
| User Datagram Protocol               | 7.63            | 7200    | 0.11          | 57600    | 29.24    | 0           | 0         | 0.00       | 7200  |
| UDP Encapsulation of IPsec Packets   | 0.47            | 448     | 0.00032       | 173      | 0.09     | 13          | 13        | 0.01       | 448   |
| Encapsulating Security Payload       | 0.42            | 395     | 0.22          | 121780   | 61.82    | 395         | 121780    | 61.82      | 395   |
| Simple Service Discovery Protocol    | 2.36            | 2225    | 1.31          | 716264   | 363.59   | 2225        | 716264    | 363.59     | 2225  |
| Session Initiation Protocol          | 0.31            | 297     | 0.34          | 187920   | 95.39    | 297         | 187920    | 95.39      | 297   |
| Routing Information Protocol         | 0.0021          | 2       | 0.00009       | 48       | 0.02     | 2           | 48        | 0.02       | 2     |
| Real-Time Transport Protocol         | 1.12            | 1060    | 0.33          | 182002   | 92.39    | 1060        | 182002    | 92.39      | 1060  |
| Real-time Transport Control Protocol | 0.0127          | 12      | 0.0015        | 820      | 0.42     | 12          | 240       | 0.12       | 24    |
| Network Time Protocol                | 0.19            | 184     | 0.0162        | 8832     | 4.48     | 184         | 8832      | 4.48       | 184   |

| Protocol                          | Percent Packets | Packets | Percent Bytes | Bytes    | Bits/s  | End Packets | End Bytes | End Bits/s | PDU's |
|-----------------------------------|-----------------|---------|---------------|----------|---------|-------------|-----------|------------|-------|
| Domain Name System                | 3.08            | 2905    | 0.55          | 301343   | 152.97  | 2905        | 301343    | 152.97     | 2905  |
| Transmission Control Protocol     | 88.04           | 83114   | 4.56          | 2494088  | 1266.04 | 71488       | 2167932   | 1100.48    | 83114 |
| Hypertext Transfer Protocol       | 10.73           | 10132   | 25.08         | 13711833 | 6960.34 | 5898        | 4044264   | 2052.93    | 10136 |
| JPEG File Interchange Format      | 1.22            | 1155    | 17.76         | 9706796  | 4927.32 | 1155        | 9706796   | 4927.32    | 1155  |
| Compuserve GIF                    | 1.20            | 1133    | 2.99          | 1635912  | 830.41  | 1133        | 1635912   | 830.41     | 1133  |
| Internet Control Message Protocol | 0.02            | 20      | 0.04          | 20636    | 10.48   | 20          | 20636     | 10.48      | 20    |
| Address Resolution Protocol       | 4.26            | 4026    | 0.21          | 112728   | 57.22   | 4026        | 112728    | 57.22      | 4026  |

Table 6. Decomposition of different packet types from capture

Based on the statistics provided in the table, the network traffic was TCP-based, accounting for 88.04% of the total packets. Wireshark TCP filters indicate substantial request-response activity involving IP addresses within the network. A notable portion of the traffic, approximately 10.73% of packets and 25.08% of total bytes, was associated with HTTP communication, suggesting significant web browsing activity. Additionally, a large amount of JPEG and GIF file transfers were observed, making up 17.76% and 2.99% of the total bytes, respectively, which implies the transmission of multimedia content. UDP traffic, particularly DNS queries (3.08% of packets), was also present, indicating active domain name resolution. This analysis is further supported by forensic verification using network monitoring tools, with relevant evidence documented in Evidence File IDs referenced in Section 5, Table 4 of this report.

| IP Address    | As Sender (Request) | As Receiver (Response) |
|---------------|---------------------|------------------------|
| 192.168.15.4  | 34554               | 38643                  |
| 192.168.1.254 | 1486                | 1496                   |
| 192.168.1.254 | 12                  | 23                     |
| 192.168.15.1  | 2154                | 0                      |
| 192.168.1.64  | 6818                | 8084                   |

Table 7. Decomposition of IP traffic (Inbound & Outbound)

It is evident from the information presented in the table that 192.168.15.4 was the most prominent player in network activity, with the highest number of packets sent and received. The activity observed on 192.168.15.1 suggests that it did not receive any packets, indicating a role as a broadcasting device or a system primarily focused on outgoing traffic. Additionally, 192.168.1.254 and 192.168.1.64 exhibited moderate network activity, while 192.168.1.254 had minimal engagement, representing an external connection or a low-traffic endpoint.

## 4. Detailed Findings

### 4.1 Important Network Players

| No | MAC Address with Name Resolution         | IP Address                                                              | Vendor                          | Device    | OS                    |
|----|------------------------------------------|-------------------------------------------------------------------------|---------------------------------|-----------|-----------------------|
| 1  | Apple_e2:c0:ce                           | 192.168.15.4                                                            | Apple, Inc.                     | Macintosh | Intel Mac OS X 10.5.4 |
| 2  | ARRISGroup_99:98:68 (Commscope_99:98:68) | 192.168.1.254                                                           | ARRIS Group, Inc.               | Router    | -                     |
| 3  | HonHaiPrecis_2e:4f:60                    | 192.168.15.1                                                            | Hon Hai Precision Ind. Co.,Ltd. | Router    | -                     |
| 4  | HonHaiPrecis_2e:4f:61                    | 192.168.1.64                                                            | Hon Hai Precision Ind. Co.,Ltd. | Router    | -                     |
| 5. | WillSelfDestruct.Com                     | 192.168.1.254(IP Address Locally Assigned By The Router To The Website) | -                               | -         | -                     |

Table 8. Important Devices in The Network (Inbound & Outbound)

To determine the IP addresses of the different MAC devices on the network, the investigator examined the source, and destination addresses on the Ethernet and IP packets being exchanged over this network. This examination was then correlated with the findings between the two layers of source and destination addresses. This revealed that devices with MAC addresses Apple\_e2:c0:ce, ARRISGroup\_99:98:68, HonHaiPrecis\_2e:4f:60, and HonHaiPrecis\_2e:4f:61 owned the IP addresses 192.168.15.4, 192.168.1.254, 192.168.15.1, and 192.168.1.64, respectively.

## 4.2 Network Structure

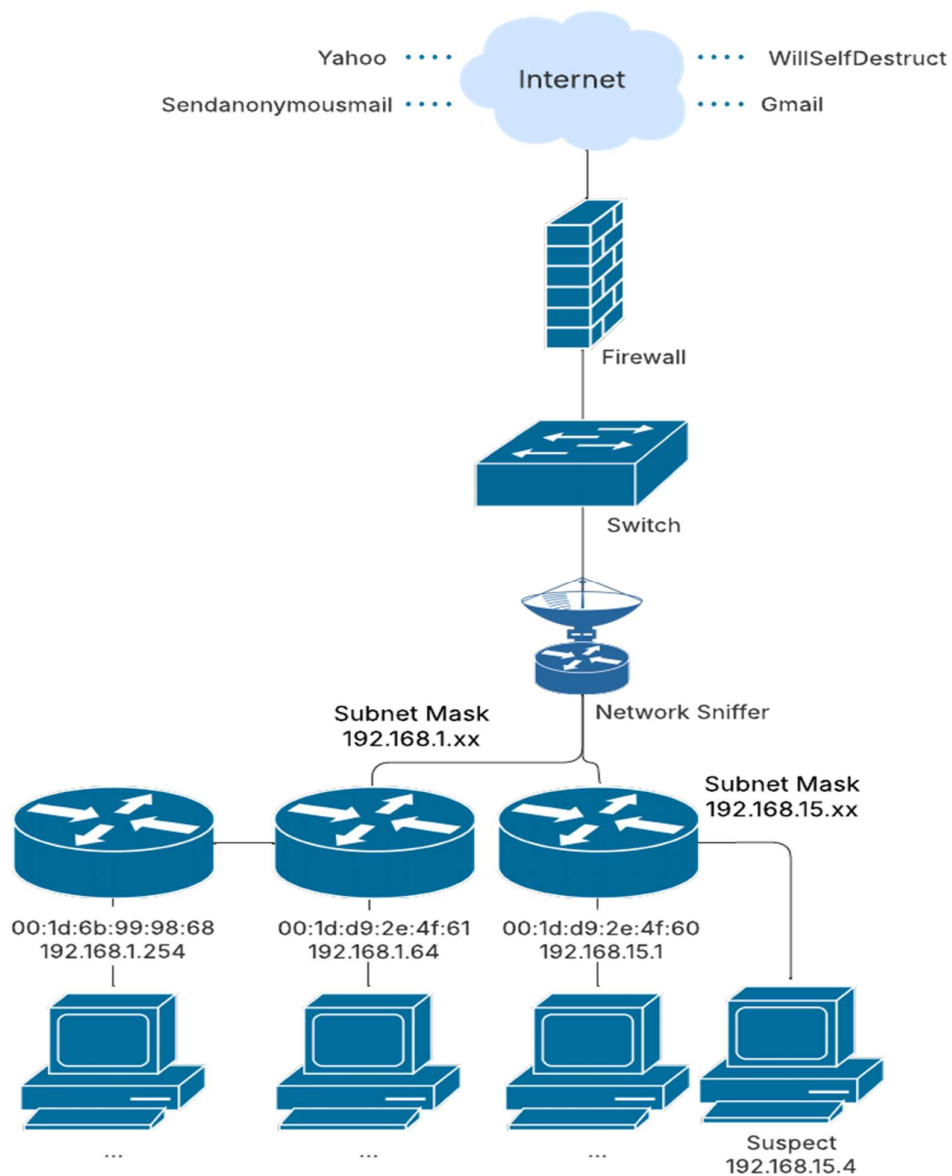


Figure 12. Possible Network Structure based on reconstruction from XYZ.pcap

The internal IP 192.168.1.1 has shown to serve as the network's DNS resolver, evidence by repeated DNS requests that was sent to this address from the internal devices like (192.168.15.4). With the use of filters like `dns && ip.dst == 192.168.1.1` it reveals queries for domains such as WillSelfDestruct.com and sendanonymousemail.net, perfectly aligns with the attackers' activities. The entire network consists of a simple format that follows the essential rules of a switch directing flow of the connection to 2 routers, 1 of which is the dorm router, and the other the campus specific one. Both of which are being monitored by the network sniffer for a period for the evidence, the router directly connected to the campus was installed by the boyfriend of one of the dorm students, but the attacker has been using the dorm router connected to their Mac Device.

### 4.3 Activity Timeline for 192.168.15.4

| Packet No. | Activity                                                                            | Destination    | Inference                                                                                   |
|------------|-------------------------------------------------------------------------------------|----------------|---------------------------------------------------------------------------------------------|
| 26211      | Google Search                                                                       | 74.125.19.99   | Searching for Dark Knight trailer                                                           |
| 26249      | Visiting news.google.com                                                            | 74.125.39.99   | Going to Google News                                                                        |
| 51933      | Google Search                                                                       | 74.125.19.103  | Searching for Sacramento tourist information                                                |
| 52018      | Visiting <a href="http://www.hellosacramento.com">www.hellosacramento.com</a>       | 65.182.192.74  | Just Visiting The Site <a href="http://www.hellosacramento.com">www.hellosacramento.com</a> |
| 72597      | Google Search                                                                       | 74.125.19.104  | Searching "how to annoy people"                                                             |
| 73157      | Visiting annoy.com                                                                  | 66.166.239.194 | Visiting Annoy.com                                                                          |
| 74059      | Google Search                                                                       | 74.125.19.104  | Searching for anonymous mail services                                                       |
| 74334      | Google Search                                                                       | 74.125.19.104  | Searching "how to harass a teacher"                                                         |
| 74920      | Yahoo Answers search query                                                          | 209.73.187.220 | Searching for legal consequences of harassment                                              |
| 75852      | Google Search                                                                       | 74.125.19.104  | Searching for Google Calendar                                                               |
| 80614      | Visiting <a href="http://www.sendanonymousemail.net">www.sendanonymousemail.net</a> | 69.80.225.91   | Sending first harassment email                                                              |
| 83601      | Visiting <a href="http://www.willselfdestruct.com">www.willselfdestruct.com</a>     | 69.25.94.22    | Sending a second harassment email                                                           |

Table 9. Activity Timeline based on network forensic analysis and event reconstruction

**Note:** Only distinct IP addresses with different time stamps are mentioned on the table. Each IP has a sequence of following TCP/HTTP packets following it which is not captured in this table. Packet numbers provided in the table indicate the first occurrence of the transaction.

## 4.4 Background Evidence

- 1) The original PCAP file was secured using different cryptographic hashing algorithms such as MD5, SHA1, and SHA256 to ensure the chain-of-custody compliance. The hashes have been cross checked multiple times against the scenarios meta data to confirm any signs of tampering or data corruption during said acquisition.
- 2) The private IP address 192.168.15.4 was found as the origin of the threats sent to Lily Tuckridge, and it correlates to the public IP 140.247.62.34 as shown in Evidence 1, this was further confirmed via MAC address mapping 00:17:f2:e2:c0:ce which was found to be an apple device being used for the attacks.
- 3) HTTP protocol dissection showed the times Willselfdestruct was accessed by different IP addresses within the network, not limiting to just those by the attacker 192.168.15.4, specifically 192.168.15.7 accessed it as well, but only 192.168.15.4 matched the corresponding MAC address. EV2.
- 4) Information on the website Willselfdestruct with consideration of the dns, limiting to 2 packets showing us MAC addresses for both the attacker 00:17:f2:e2:c0:ce and the campus router 00:1c:b3:9a:4e:1a this was seen due to the unencrypted TCP handshakes being exposed. EV3.
- 5) Limiting the MAC addresses to just one (the attacker) and searching for relations with the website will self-destruct, MAC address 00:17:f2:e2:c0:ce being the attackers specific device. This confirmed the use of the apple hardware, and the number of packets is something to note. EV4.
- 6) Showing all said threats sent to lily, one being the original message sent through sendanonymousmail.net payload containing "Stop teaching. Start running.," and the other from willselfdestruct.com the self-deleting note reads "You can't find us." Packet 80614, and 83601, respectively. EV5.
- 7) By filtering based on MAC Address we find all searches done with `eth.addr == 00:17:f2:e2:c0:ce && frame contains "search_result"`, all of which can be accessed using the file EV6.
- 8) Under that filter we find the packet 74920 which searches the following "can I go to jail for harassing my teacher", again the exact MAC address we found previously, and with timestamps aligning to the harassment event. EV7.
- 9) Contains all the email logins done within the network during the packet sniffing allocated time, including the user [elishavet@gmail.com](mailto:elishavet@gmail.com) and the attacker [jcoachj@gmail.com](mailto:jcoachj@gmail.com) EV8.
- 10) Using said previous filter we can find [elishevet@gmail.com](mailto:elishevet@gmail.com) with its relevant cookie pair making use of the boyfriend's router, which was installed into the dorm, this was deduced by the different destination being CommScope EV9.
- 11) We can now find all the mail logons done from the specific mac address of the attacker `eth.addr == 00:17:f2:e2:c0:ce && frame contains "GET /mail"` EV10.

- 12) Using that we can find jcoachj@gmail.com being accessed authenticated from that specific Mac Address 00:17:f2:e2:c0:ce in packet 78967 EV11.
- 13) After finding out that "jcoachj" belongs to the attacker we can find that there's 130~ different packets including Gmail, and Google drive activity from specific email using http.cookie\_pair contains "jcoachj" EV12.
- 14) Now finally using eth.addr == 00:17:f2:e2:c0:ce && http.cookie\_pair contains "jcoachj" we can test whether the number of packets sent matches that of which sent using the apple device (MAC address) EV13.
- 15) Now considering a different application we used Network Miner to get more information to further help us with justifying the culprit, by searching keywords in the application such as "Teacher" and "Gmail.com" reveals frame number 74920, again mentioning the "can I go to jail for harassing my teacher", and 79715 which mentions the cleartext cookie of jcoachj tied to the ip address 192.168.15.4.(Figure 10)
- 16) Considering that Wireshark provides a more in-depth analysis, network miner still provided a clear and understanding approach, for example the messages tab directly highlighted both the sendanonymousemail.net and Willselfdestruct.com activities with its corresponding frame numbers. This also confirmed the attacker's activity clustered within a 15-minute window which predated the email deliveries.(Figure 9)
- 17) The attacker's User-Agent was shown multiple times with the help of Network miner, which read Mozilla/5.0 (Macintosh; PPC Mac OS X) indicating an outdated macOS software, with many vulnerabilities.
- 18) Another thing to consider is the username amy789smith was detected in packet 90471 via Yahoo messenger authentication which was sharing the same IP and MAC address as jcoachj, potentially device sharing between the 2 individuals.
- 19) Considering that it's concluded that the suspect used the email [jcoachj@gmail.com](mailto:jcoachj@gmail.com) from the list of suspects provided it matches with Johnny Coach a student from the Chemistry lecture that Lily Tuckridge was teaching.

## Supporting Evidence Presented

| Evidence Identifier | Content                                     | Content Source           | Filename                                       | Hashes/Checksums                             |                                                          |                                                                                              |
|---------------------|---------------------------------------------|--------------------------|------------------------------------------------|----------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------|
|                     |                                             |                          |                                                | MD5                                          | SHA1                                                     | SHA256                                                                                       |
| 1                   | Private IP Proof                            | Wireshark Version 4.4.3  | ev_1_privateipproof.txt                        | e188a1928<br>8369b8b4a<br>84edd0995<br>398e3 | 9f27d994c<br>c8b758932<br>c4ad45923<br>7baf65bac6<br>5aa | dd8b4e192<br>48b298f5d<br>d53fcfa061<br>1c8761582<br>0b301676a<br>fe9923991<br>d6937e1cd     |
| 2                   | Will Self-Destruct Packet Capture           | Wireshark Version 4.4.3  | ev_2_Willselfdestruct.pcap                     | 0babbe4f4<br>13c62dd80<br>564925714<br>dc31e | 98023a060<br>0ca97ff943<br>192fb0963<br>4a985900e<br>6df | 84ca4ac37f<br>b42bd401c<br>bac9dcfc45<br>d2e85c6d3<br>e5814dfece<br>52cfb52cd<br>ebfccc7     |
| 3                   | Will Self-Destruct with DNS Info            | Wireshark Version 4.4.37 | ev_3_willselfdestructwiththednsinfo.txt        | a6683caa7<br>4a9428b8a<br>7b41f329f<br>6a014 | eeeff830e0<br>83f60a3acc<br>cd8030a14<br>9f78b7c2ce<br>2 | 956e22b24<br>669232c5c<br>38f4f2671<br>6406d7baa<br>e5a4350a9<br>cdfe93fba0<br>86d4ac031     |
| 4                   | Specific MAC Address for Will Self-Destruct | Wireshark Version 4.4.3  | ev_4_specificmacaddressforwillselfdestruct.txt | c2c16a557<br>433ce1080<br>0573937d8<br>59396 | ec2c1b0e9<br>bef6adb33<br>dba1b3406<br>2dfe6ad86<br>630e | 98ab5c2a6<br>525a8c8ba<br>92475efce<br>db3304186<br>8134f8254<br>2731db643<br>91d0fd625<br>0 |
| 5                   | All Threats Sent to Lilly                   | Wireshark Version 4.4.3  | ev_5_allthethreatsentsolly.pcap                | 13f4e50cef<br>f2b30a9b5f<br>314681bcc<br>e29 | bbc70ce15<br>af18c5e7dc<br>6bdb9a82d<br>f9633d13d<br>b8d | 27bcd2afaf<br>9c938e8fb<br>9d9fb476b<br>75f98bd4e<br>a43e710e7<br>47e9c62ecf<br>e9e5e5df     |
| 6                   | Searches Done by MAC Address                | Wireshark Version 4.4.3  | ev_6_searchesdonebymacaddress.pcap             | 87f2cef148<br>b663f8f32<br>18e7fe5ca8<br>e2e | 0042a562b<br>8a2055233<br>9c197850d<br>a52dd3ee0<br>bbc2 | ce0fd65fff<br>15f776345<br>2ea73bdc5<br>40c89cc64<br>4dc529052                               |

|    |                                                         |                               |                                              |                                              |                                                          |                                                                                              |
|----|---------------------------------------------------------|-------------------------------|----------------------------------------------|----------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------|
|    |                                                         |                               |                                              |                                              |                                                          | f58bdc34<br>8d66c7a85                                                                        |
| 7  | Packet<br>74920<br>Incrimination                        | Wireshark<br>Version<br>4.4.3 | ev_7_packet74920incrimination.txt            | df8f71207f<br>fbd6e0c08<br>7f71d4a26<br>e868 | b0e0205f6<br>84909a817<br>a83041481<br>eeb761018<br>e267 | 9d432f7f5<br>133a6872d<br>c62007afc<br>3aa4e4c51<br>0783faed6<br>1d700729f<br>2d435ab5b<br>a |
| 8  | All the<br>Logins of<br>the Emails<br>in the<br>Network | Wireshark<br>Version<br>4.4.3 | ev_8_alltheloginsofthemailsinthenetwork.pcap | 23ca932a1<br>3dbf9abe5<br>3f6c77f40b<br>768a | 271cf61f72<br>627044e78<br>63baac19f5<br>cbe44b632<br>38 | cdf5b7840<br>7f87dec1<br>948ac33e9<br>94f83f0b5<br>192864f63<br>bd4bccb9a<br>12ed22ea3<br>0  |
| 9  | "Elishevet<br>My Goat"<br>Note                          | Wireshark<br>Version<br>4.4.3 | ev_9_elishevetinformation.txt                | b213f46e0<br>1f78e507c<br>133d99e96<br>f7b39 | fa450908b<br>b2aa7658e<br>853737a0e<br>097ebe383<br>4b14 | 06f4c7b51<br>825e987f4<br>30f9a943a<br>59725368f<br>b4a7d4eab<br>c66d273c1<br>9029cf86a<br>8 |
| 10 | All Mail<br>Logons<br>from MAC<br>Address               | Wireshark<br>Version<br>4.4.3 | ev_10_allmaillogonsfromthemacaddress.pcap    | 311dae2c8<br>0487376c3<br>b23881907<br>19ee2 | cc32bf823<br>dfe5eb980f<br>c78881b50<br>21ec4fbb4<br>4dd | 42fab1319f<br>112da5902<br>8b473b605<br>c7406ce62<br>78c394b9b<br>0517fdeb4<br>6be0736ef     |
| 11 | JCoach<br>Email Stiff<br>from MAC<br>Address            | Wireshark<br>Version<br>4.4.3 | ev_11_jcoachemailstifffrommacaddress.txt     | cb5487608<br>dff701a1e0<br>c0d96dd60<br>8fa2 | 3a74e3a14<br>b69923968<br>9e2e63dc2<br>ba2e96f99<br>37ec | b3e539f2e<br>6cfccc3d41<br>7a4c9ddcd<br>762f47e2a<br>63b0d9aa7<br>47c96479a<br>b1e233a05     |
| 12 | JCoach<br>Cookie<br>Pairs                               | Wireshark<br>Version<br>4.4.3 | ev_12_jcoachjcookiepairs.pcap                | bd69bd8ed<br>c339f54bd<br>e4d50bf8a<br>4f775 | b95c7c33e<br>9bf82161d<br>a90ccc3f54<br>dcaff52e46<br>13 | 085321d65<br>562936aad<br>d5ca2ffb34<br>f05229b61<br>8a91c2f69<br>ad2a8984e<br>ec0757675     |

|    |                                             |                               |                                                |                                              |                                                          |                                                                                              |
|----|---------------------------------------------|-------------------------------|------------------------------------------------|----------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------|
| 13 | JCoach<br>Comparison<br>with MAC<br>Address | Wireshark<br>Version<br>4.4.3 | ev_13_jcoachjcompariso<br>nwithmacaddress.pcap | bd69bd8ed<br>c339f54bd<br>e4d50bf8a<br>4f775 | b95c7c33e<br>9bf82161d<br>a90ccc3f54<br>dcaff52e46<br>13 | 085321d65<br>562936aad<br>d5ca2ffb34<br>f05229b61<br>8a91c2f69<br>ad2a8984e<br>ec0757675     |
| 14 | TCP<br>Stream<br>Proofing                   | Wireshark<br>Version<br>4.4.3 | ev_14_TCPStreamProofi<br>ng.txt                | 813e5df9b<br>a06b9c52c<br>280a3526c<br>f1a2e | 95b7fa833<br>a1e00fe96e<br>ce1570adfd<br>b91cdcded<br>42 | c56595987<br>c85de6e67<br>51441a153<br>3f592ab68<br>b743203ca<br>13bd7419e<br>6e28ca882<br>8 |

Table 4. Tabulated list of evidence supporting the Forensic report

## 5. Conclusions

The forensic analysis of the XYZ.pcap file identifies that Johnny Coach who owns the email [jcoachj@gmail.com](mailto:jcoachj@gmail.com) is the perpetrator behind the harassment targeting the teacher Lily Tuckridge, a Chemistry instructor. The investigation was conducted using Wireshark and NetworkMiner, revealing that the attacker's device (MAC 00:17:f2:e2:c0:ce, IP 192.168.15.4), sent harassing emails via the use of services such as sendanonymousmail.net, and willselfdestruct.com with the messages such as "Stop teaching. Start running," and "You can't find us." These emails were routed through the dorm router (00:1d:d9:2e:4f:60), with the help of the network sniffer we were able to go over the packets flowing through the network and come up with the following conclusion. The MAC address 00:17:f2:e2:c0:ce, tied to an Apple device, was consistently linked to jcoachj@gmail.com through clear text HTTP cookies (e.g., gmailchat=jcoachj@gmail.com/945167) found in Google service logins which can be further found in the evidence provided, by following through the TCP stream of the MAC address of the suspect, we found the Mozilla/5.0 (Macintosh; U; Intel Mac OS X 10\_5\_4; en-us) to be the main device used by Johnny Coach, and one other person that has accessed this device but not involved in the harassment. Knowing this Lily Tuckridge is legally capable of acting against the student for these threats if she chooses, furthermore, all evidence provided complies with the standards for chain of custody, and considering the best practices. All provided evidence have been encrypted with secure hashing algorithms and that will conclude the report.

## 6. Appendix A – List of figures

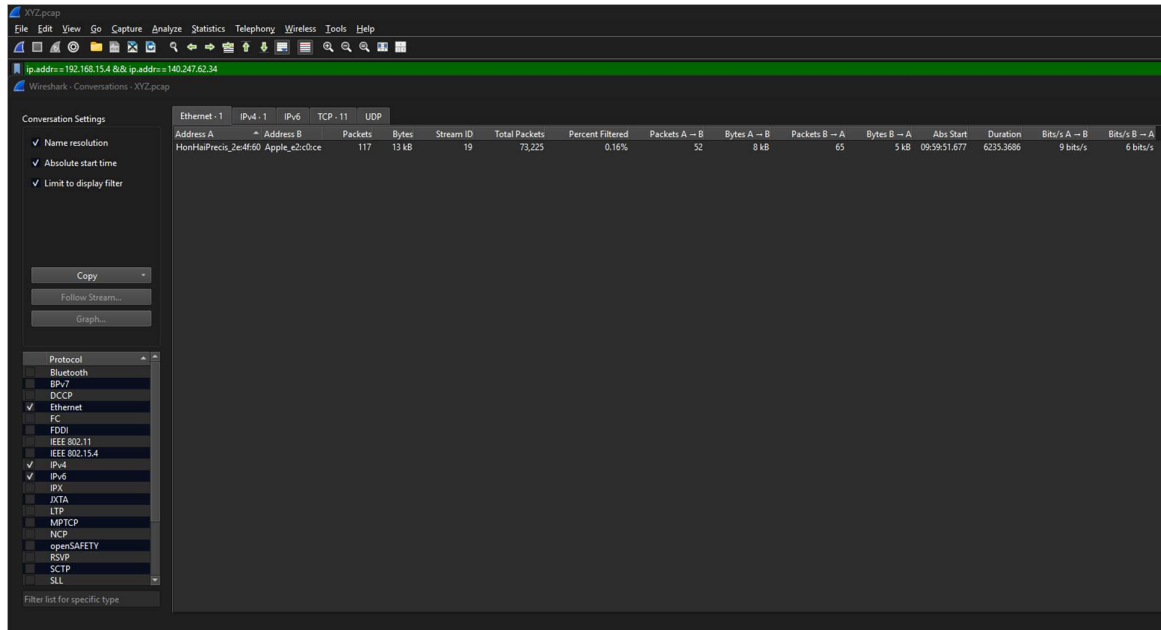


Figure 13

Description: Filtering The Data Packets Where the Public Ip of The Website and Private Ip of The Suspect Is Connected and Find the Router Through Which They Connected Using the Statistics Tab in Wireshark Tool.

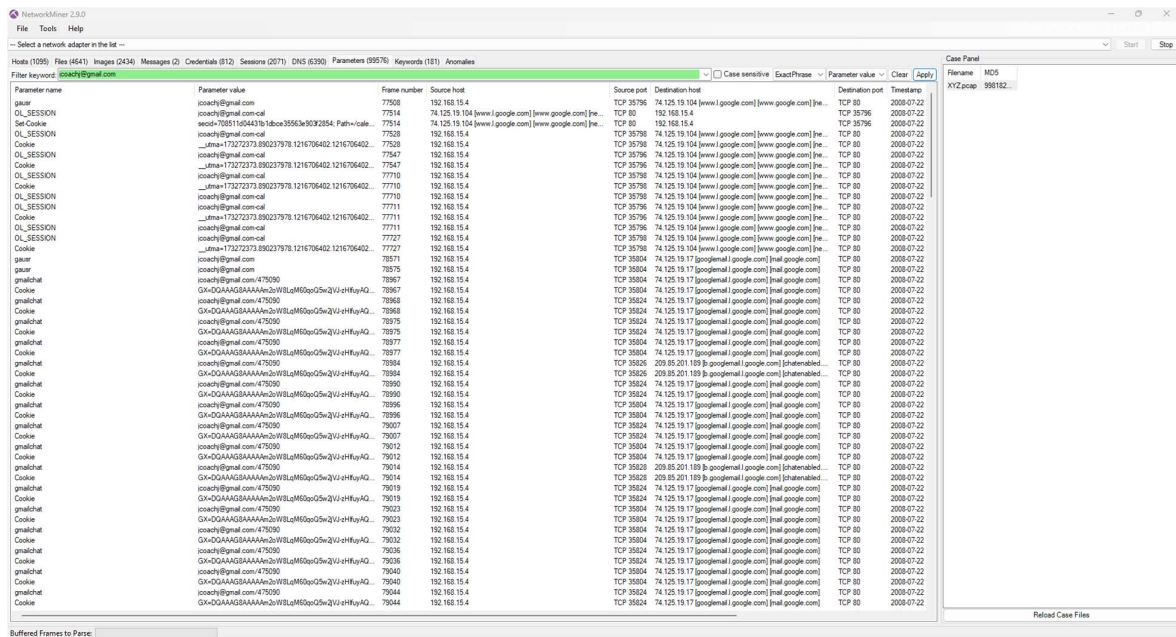


Figure 14

Description: Filtering [jcoachj@gmail.com](mailto:jcoachj@gmail.com) cookie packets

**Description:** Filtering Data Packets Related to Suspects IP Address Using Network Miner

## 7. Self-Review

Name: S.D.Nipunanga  
ID: 8145519 | 25367

I'm S.D. Nipunanga and I contributed to the report by gathering evidence in both Wireshark v4.4.3 and Network Miner v2.9.0 using a window 11 system. The evidence placed in the background evidence, and relevant appendixes were gathered. I also contributed by completing the executive summary, conclusion, steps involved and the network diagram alongside the description. The analysis process, including the review procedure, is primarily conducted as a team, as it cannot be completed by a single individual

Name: L.D.Nithila  
ID: 8145692 | 29919

I am Dulan Liynarachchi and in this report, I conducted a thorough network analysis using Wireshark Version 4.2, covering key aspects such as network capture file details, identified network components, and handling data efficiently. I examined important network players and provided a detailed activity timeline for 192.168.15.4, ensuring a structured investigation. My findings were supported with relevant evidence, and I included an appendix listing all figures for clarity. Overall, the report presents a comprehensive analysis, demonstrating my ability to utilize network monitoring tools effectively and interpret network traffic data accurately. Most of the things in the analyzation is done as a team, including the review procedure because this can't be done by a single individual.